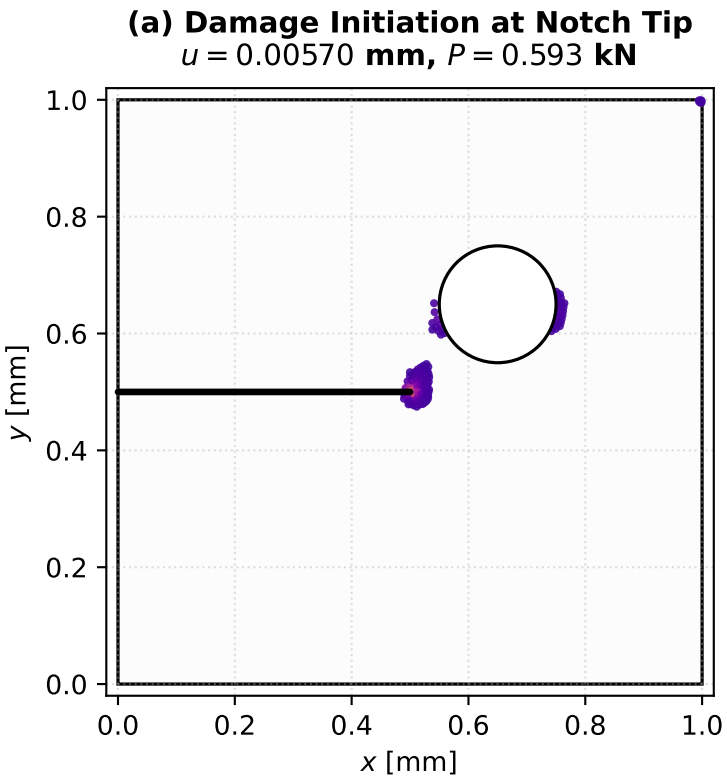
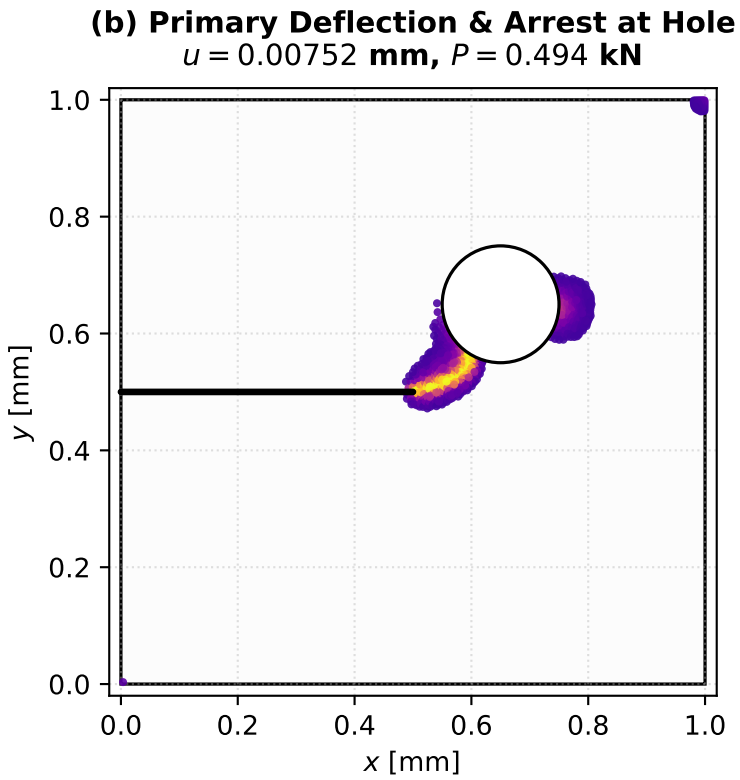
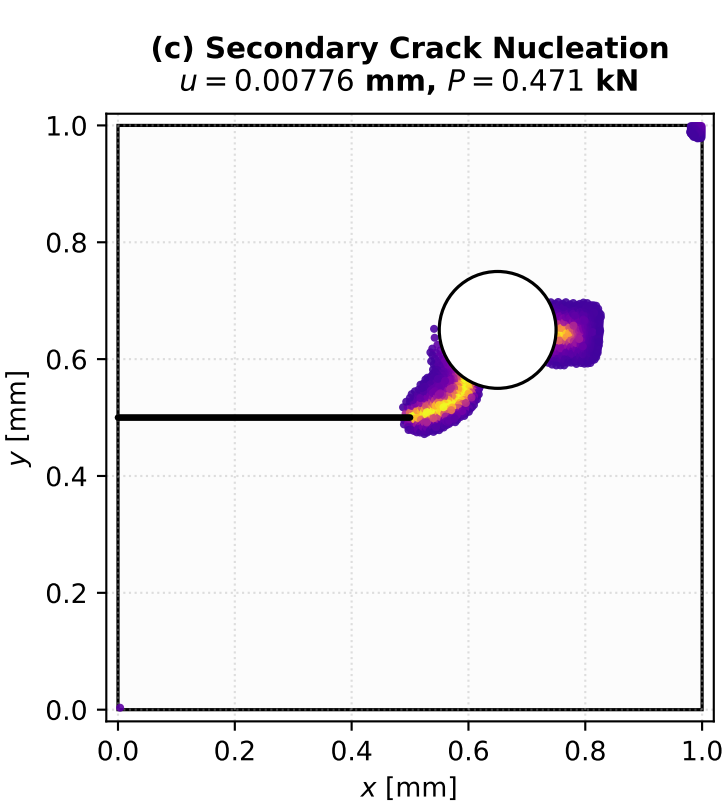
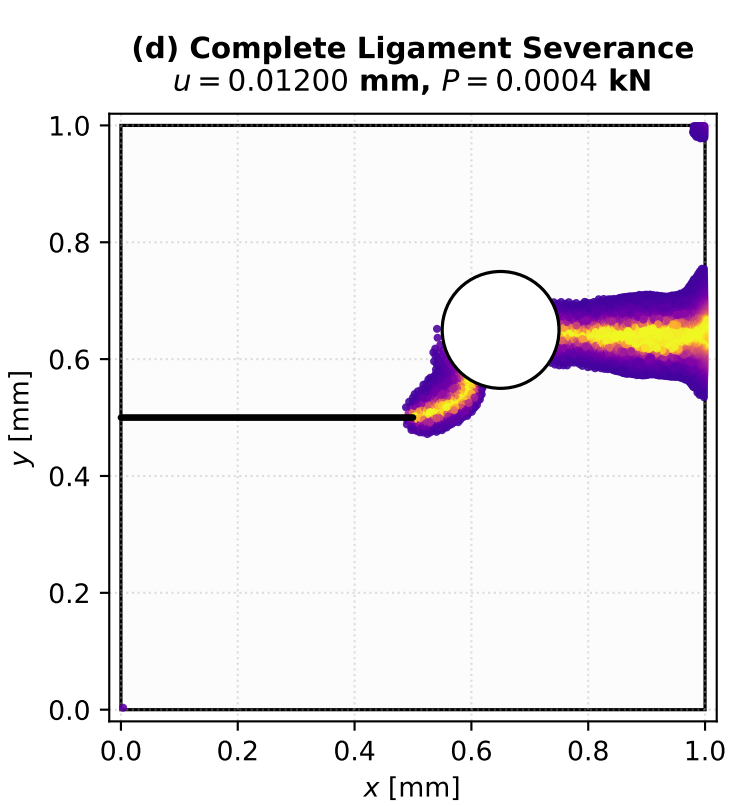
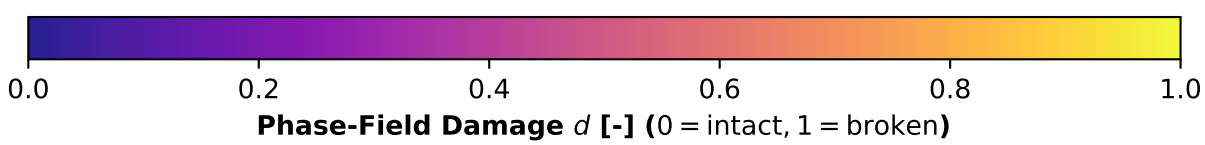
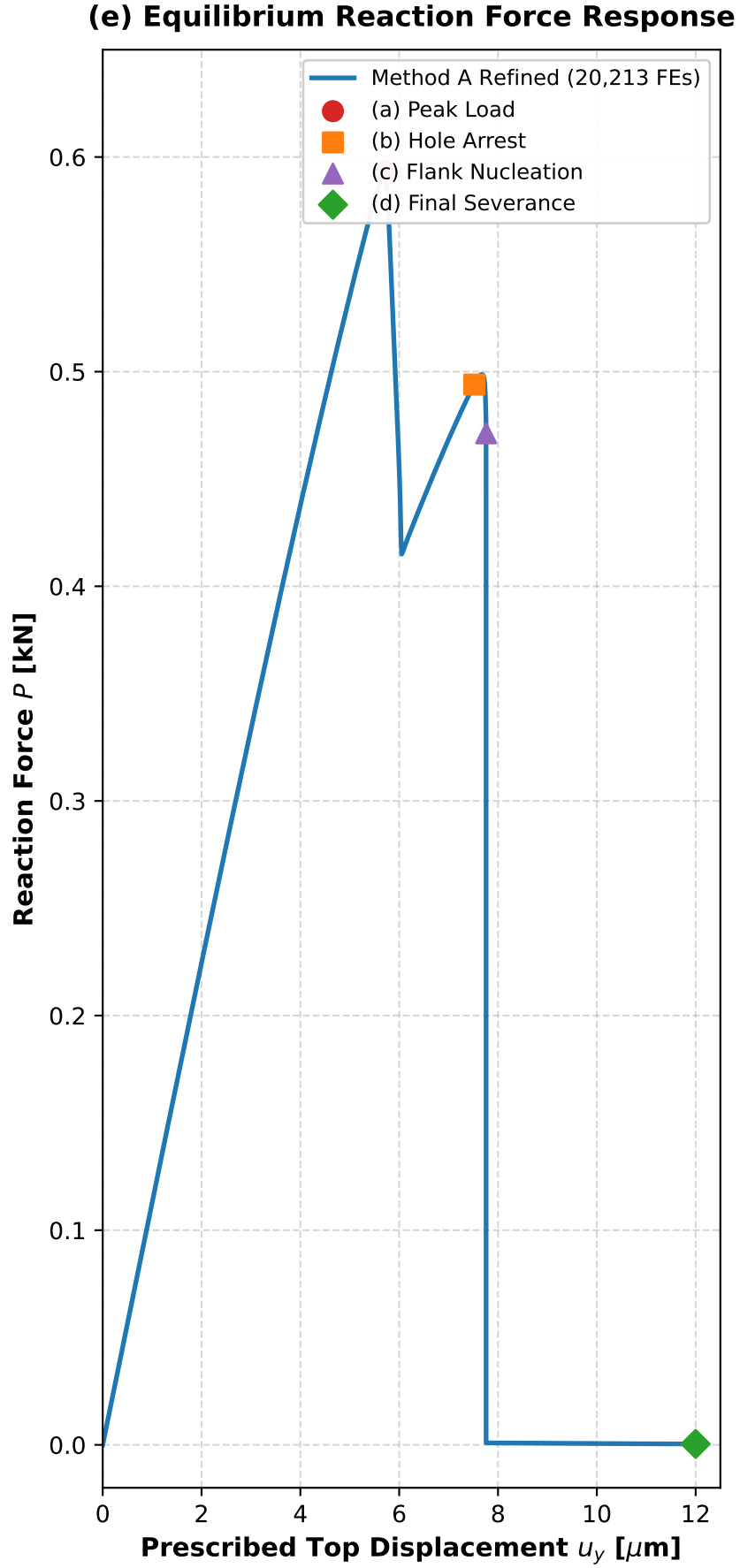

(a) Damage Initiation at Notch Tip
$u = 0.00570$ **mm,** $P = 0.593$ **kN**
y [mm]
x [mm]


(b) Primary Deflection & Arrest at Hole
$u = 0.00752$ **mm,** $P = 0.494$ **kN**
y [mm]
x [mm]


(c) Secondary Crack Nucleation
$u = 0.00776$ **mm,** $P = 0.471$ **kN**
y [mm]
x [mm]


(d) Complete Ligament Severance
$u = 0.01200$ **mm,** $P = 0.0004$ **kN**
y [mm]
x [mm]


Phase-Field Damage d [-] (0 = intact, 1 = broken)


(e) Equilibrium Reaction Force Response
Method A Refined (20,213 FEs)
(a) Peak Load
(b) Hole Arrest
(c) Flank Nucleation
(d) Final Severance
Reaction Force P [kN]
Prescribed Top Displacement u_y [μm]